THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

ANSWER KEY & MARKING SCHEME — MATHEMATICS · CLASS 8 · SET 05

Max. Marks: 60 · The Mastermind Coaching Classes, Jaipur · Alternate correct methods to be given full credit

SECTION A — Multiple Choice Questions (12 × 1 = 12) — **1 questions · 12 marks**

1. Question 1 consists of multiple-choice questions (i–xii) of 1 mark each. Choose the correct option.

(i) Which of the following is a perfect square?

(a) 1200 (b) 1296 (c) 1250 (d) 1300

Ans: (b) $36^2 = 1296$.

(ii) The cube root of −343 is:

(a) 7 (b) −49 (c) −7 (d) −3.5

Ans: (c) $(-7)^3 = -343$.

(iii) The units digit of 17^2 is:

(a) 9 (b) 7 (c) 1 (d) 4

Ans: (a) $17^2 = 289$.

(iv) The value of $(-2)^{-3}$ is:

(a) −1/8 (b) 1/8 (c) −8 (d) 8

Ans: (a) $(-2)^{-3} = 1/(-2)^3 = -1/8$.

(v) 349 in Roman numerals is:

(a) CCCIL (b) CCCXLIX (c) CCXXXXIX (d) CCCXXXXIX

Ans: (b) 300 = CCC, 40 = XL, 9 = IX.

(vi) $(1010)_2$ in decimal is:

(a) 10 (b) 8 (c) 12 (d) 5

Ans: (a) 8 + 0 + 2 + 0 = 10.

(vii) A quadrilateral whose diagonals are equal and bisect each other at right angles is a:

(a) rectangle (b) rhombus (c) kite (d) square

Ans: (d) Only a square has all these properties together.

(viii) The diagonals of a kite meet at:

(a) 60° (b) 45° (c) any angle (d) right angles

Ans: (d) The diagonals of a kite are perpendicular.

(ix) $(x - 5)(x + 5)$ equals:

(a) $x^2 + 25$ (b) $x^2 - 10x + 25$ (c) $x^2 - 25$ (d) $x^2 - 10$

Ans: (c) $(a - b)(a + b) = a^2 - b^2$.

(x) If $x + 1/x = 3$, then $x^2 + 1/x^2$ equals:

(a) 9 (b) 11 (c) 7 (d) 5

Ans: (c) 9 − 2 = 7.

(xi) If 4 : 7 = 20 : x, then x is:

(a) 28 (b) 32 (c) 40 (d) 35

> **Ans: (d)** $4x = 140 \Rightarrow x = 35$.

(xii) 6 men build a wall in 10 days. 15 men will build it in:

(a) 4 days (b) 25 days (c) 6 days (d) 9 days

> **Ans: (a)** Inverse variation: $6 \times 10 = 15 \times d \Rightarrow d = 4$.

SECTION B — Very Short Answer (6 × 2 = 12) — 6 questions · 12 marks

Questions 2 to 7 carry 2 marks each.

2. Find the square root of 1521 by prime factorisation.

> $1521 = 3^2 \times 13^2$ (1 mark)
> $\sqrt{1521} = 3 \times 13 = 39$ (1 mark)

3. Find the smallest number by which 500 must be multiplied to get a perfect cube.

> $500 = 2^2 \times 5^3$ (1 mark)
> Multiply by 2: $1000 = 10^3$ (1 mark)

4. Write 0.0000081 in standard form and 7.5×10^4 in usual form.

> 8.1×10^{-6} (1 mark)
> 75000 (1 mark)

5. Write the Roman numeral MCDXLVII in the Hindu-Arabic system.

> 1000 + 400 + 40 + 7 (1 mark) = 1447 (1 mark)

OR

Convert $(321)_4$ into a decimal number.

> 3×16 + 2×4 + 1 (1 mark) = 57 (1 mark)

6. Two adjacent angles of a parallelogram are in the ratio 1 : 4. Find them.

> Sum = 180° ⇒ 5 parts = 180°, 1 part = 36° (1 mark)
> Angles: 36° and 144° (1 mark)

7. Expand $(2a - 3)^2$ and $(x + 7)(x + 3)$.

> $(2a - 3)^2 = 4a^2 - 12a + 9$ (1 mark)
> $(x + 7)(x + 3) = x^2 + 10x + 21$ (1 mark)

SECTION C — Short Answer (3 × 3 = 9) — 3 questions · 9 marks

Questions 8 to 10 carry 3 marks each.

8. Find the cube root of −1.331 and the square root of 0.0625.

> $\sqrt[3]{(-1.331)} = -\sqrt[3]{(1331/1000)} = -11/10 = -1.1$ (1½ marks)
> $\sqrt{0.0625} = \sqrt{(625/10000)} = 25/100 = 0.25$ (1½ marks)

9. The angles of a quadrilateral are x°, (x + 10)°, (x + 20)° and (x + 30)°. Find each angle.

> $4x + 60 = 360$ (1 mark) ⇒ $x = 75$ (1 mark)
> Angles: 75°, 85°, 95°, 105° (1 mark)

OR

The diagonals of rectangle ABCD meet at O. If OA = 5 cm and ∠AOB = 60°, find BD and AB.

> Diagonals are equal and bisect each other: OB = OA = 5 cm, BD = 10 cm (1½ marks)
> ΔOAB has OA = OB and ∠AOB = 60°, so it is equilateral ⇒ AB = 5 cm (1½ marks)

10. Using identities, find: (i) 103^2 (ii) 59×61 (iii) $(2.1)^2$

> (i) $(100 + 3)^2 = 10000 + 600 + 9 = 10609$ (1 mark)
> (ii) $(60 - 1)(60 + 1) = 3600 - 1 = 3599$ (1 mark)
> (iii) $(2 + 0.1)^2 = 4 + 0.4 + 0.01 = 4.41$ (1 mark)

SECTION D — Long Answer (3 × 5 = 15) — 3 questions · 15 marks

Questions 11 to 13 carry 5 marks each.

11. (i) Find the smallest number by which 2700 must be divided so that the quotient is a perfect square. Also find the square root of the quotient.
(ii) Find the smallest number to be added to 1000 to make it a perfect square.

> (i) $2700 = 2^2 \times 3^3 \times 5^2$ (1 mark); divide by 3 (1 mark); quotient 900, $\sqrt{900} = 30$ (1 mark)
> (ii) $31^2 = 961 < 1000 < 32^2 = 1024$ (1 mark); add $1024 - 1000 = 24$ (1 mark)

OR

(i) Find the cube root of 17576 by prime factorisation.
(ii) A cubical box has volume 5.832 m^3. Find the length of its edge.

> (i) $17576 = 2^3 \times 13^3$ (2 marks); cube root $= 2 \times 13 = 26$ (1 mark)
> (ii) $5.832 = 5832/1000 = (18/10)^3 \Rightarrow$ edge $= 1.8$ m (2 marks)

12. Simplify: (i) $3^{-1} + 5^{-1}$ (ii) $(2^5 \times 2^{-2}) \div 2^{-1}$ (iii) $(-5)^0 \times 5^{-2}$ (iv) Write 0.00000052 in standard form (v) Find x if $2^x \div 2^{-3} = 2^8$

> (i) $1/3 + 1/5 = 8/15$ (ii) $2^3 \div 2^{-1} = 2^4 = 16$ (iii) $1 \times 1/25 = 1/25$ (iv) 5.2×10^{-7} (v) $x + 3 = 8 \Rightarrow x = 5$ (1 mark each)

OR

(i) Write 1994 in Roman numerals. (ii) Convert 77 to base 2. (iii) Add $(101)_2$ and $(11)_2$. (iv) Convert $(143)_5$ to decimal. (v) Convert 100 to base 3.

> (i) MCMXCIV (ii) $77 = 64 + 8 + 4 + 1 = (1001101)_2$ (iii) $5 + 3 = 8 = (1000)_2$ (iv) $25 + 20 + 3 = 48$ (v) $100 = 81 + 18 + 1 = (10201)_3$ (1 mark each)

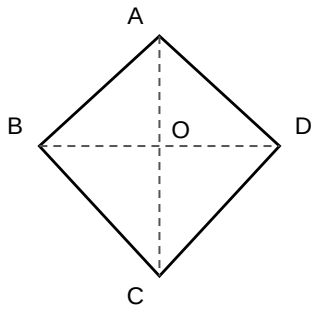
13. (i) Divide ₹ 5,400 among three persons in the ratio 2 : 3 : 4.
(ii) If 12 notebooks cost ₹ 180, find the cost of 20 notebooks.
(iii) 3 pumps empty a tank in 8 hours. How long will 4 pumps take?
(iv) Find x if 5 : 8 :: x : 56.
(v) The ages of two brothers are in the ratio 4 : 5 and their sum is 45 years. Find their ages.

> (i) 9 parts, 1 part = ₹ 600 ⇒ ₹ 1200, ₹ 1800, ₹ 2400 (1 mark)
> (ii) 1 notebook = ₹ 15 ⇒ ₹ 300 (1 mark)
> (iii) Inverse: $3 \times 8 = 4 \times t \Rightarrow t = 6$ hours (1 mark)
> (iv) $8x = 280 \Rightarrow x = 35$ (1 mark)
> (v) 9 parts = 45 ⇒ 20 and 25 years (1 mark)

SECTION E — Case-Based Questions (3 × 4 = 12) — 3 questions · 12 marks

Questions 14 to 16 carry 4 marks each (1 + 1 + 2).

14. A kite-shaped decoration ABCD is cut from coloured paper. AB = AD = 5 cm, CB = CD = 9 cm, ∠BAD = 70° and ∠ABC = ∠ADC = 100°. The diagonals AC and BD meet at O.



(i) Which two angles of the kite are equal?

∠ABC = ∠ADC (= 100°)

(ii) Find the perimeter of the decoration.

2(5 + 9) = 28 cm

(iii) Find ∠BCD.

Angle sum 360°: ∠C = 360° − (70° + 100° + 100°) (1 mark) = 90° (1 mark)

OR

(iii) Find ∠AOB and ∠ABO.

Diagonals of a kite are perpendicular ⇒ ∠AOB = 90° (1 mark); AC bisects ∠A ⇒ ∠BAO = 35°, so ∠ABO = 90° − 35° = 55° (1 mark)

15. A square garden has an area of 2025 m². The owner wants to fence it on all four sides.

(i) Find the side of the garden.

√2025 = 45 m

(ii) Find the length of fencing required.

4 × 45 = 180 m

(iii) Find the cost of fencing at ₹ 120 per metre.

180 × 120 (1 mark) = ₹ 21,600 (1 mark)

OR

(iii) The side is increased by 4 m. Find the area of the new square garden.

New side = 49 m (1 mark); area = 49² = 2401 m² (1 mark)

16. A rectangular hall has length (x + 7) m and width (x + 3) m, where x is a number of metres chosen by the architect. The floor is to be carpeted.

(i) Write the area of the hall as a polynomial in x.

$(x + 7)(x + 3) = x^2 + 10x + 21$

(ii) Find the area when x = 5.

25 + 50 + 21 = 96 m² (dimensions 12 m × 8 m)

(iii) Find the cost of carpeting at ₹ 150 per m² when x = 5.

Area = 96 m² (1 mark); cost = 96 × 150 = ₹ 14,400 (1 mark)

OR

(iii) Show that $(x + 7)(x + 3) = (x + 5)^2 − 4$.

$(x + 5)^2 − 4 = x^2 + 10x + 25 − 4$ (1 mark) $= x^2 + 10x + 21 = (x + 7)(x + 3)$ (1 mark)